STAT 4194 Project

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Packages Used

Data

Model Fits

AME Model

For the AME model, we only use the sociality and multiplicative effects, since there are no dyadic or nodal attributes for the characters.

AME Model Type

We also used a fixed rank nomination model, to fit for the count adjacency data (# of scenes shared). For example

```
           CAMERON PATRICK TREVOR MICHAEL MR CHAPIN BRUCE DEREK JOCK MRS
BLAISE
CAMERON         NA       6      0       8         0     0     1    0
0
PATRICK          6      NA      0       4         1     1     0    0
1
TREVOR           0       0     NA       0         0     0     0    0
0
MICHAEL          8       4      0      NA         0     0     1    0
0
MR CHAPIN        0       1      0       0        NA     0     0    0
0
BRUCE            0       1      0       0         0    NA     0    0
0
DEREK            1       0      0       1         0     0    NA    0
1
JOCK             0       0      0       0         0     0     0   NA
0
MRS BLAISE       0       1      0       0         0     0     1    0
NA
RIDER            0       0      0       0         0     0     0    0
```

```
0
JOEY            3       5      0       2         0     0     2    1
2
CLEM            0       0      0       0         0     0     1    0
1
KAT             2      13      0       2         1     1     0    1
3
TEACHER         0       0      0       0         0     0     0    0
0
MANDELLA        1       1      1       3         0     1     0    0
0
COWBOY          1       0      0       0         0     0     1    0
0
BIANCA          8       2      0       1         0     0     1    1
0
CHASTITY        2       0      0       1         0     0     0    0
0
WALTER          2       1      0       0         0     0     0    0
0
SHARON          1       0      0       0         0     0     0    0
0
MISS PERKY      2       3      0       1         0     0     0    0
0
           RIDER JOEY CLEM KAT TEACHER MANDELLA COWBOY BIANCA CHASTITY WALTER
CAMERON        0    3    0   2       0        1      1      8        2      2
PATRICK        0    5    0  13       0        1      0      2        0      1
TREVOR         0    0    0   0       0        1      0      0        0      0
MICHAEL        0    2    0   2       0        3      0      1        1      0
MR CHAPIN      0    0    0   1       0        0      0      0        0      0
BRUCE          0    0    0   1       0        1      0      0        0      0
DEREK          0    2    1   0       0        0      1      1        0      0
JOCK           0    1    0   1       0        0      0      1        0      0
MRS BLAISE     0    2    1   3       0        0      0      0        0      0
RIDER         NA    0    0   1       0        0      0      0        0      0
JOEY           0   NA    1   4       0        1      1      5        1      1
CLEM           0    1   NA   0       0        0      0      0        0      0
KAT            1    4    0  NA       1        6      0     11        1      4
TEACHER        0    0    0   1      NA        1      0      1        0      0
MANDELLA       0    1    0   6       1       NA      0      0        0      0
COWBOY         0    1    0   0       0        0     NA      1        0      0
BIANCA         0    5    0  11       1        0      1     NA        5      4
CHASTITY       0    1    0   1       0        0      0      5       NA      0
WALTER         0    1    0   4       0        0      0      4        0     NA
SHARON         0    1    0   2       0        0      0      3        0      3
MISS PERKY     0    0    0   2       0        0      0      1        0      0
           SHARON MISS PERKY
CAMERON         1          2
PATRICK         0          3
```

```
TREVOR          0          0
MICHAEL         0          1
MR CHAPIN       0          0
BRUCE           0          0
DEREK           0          0
JOCK            0          0
MRS BLAISE      0          0
RIDER           0          0
JOEY            1          0
CLEM            0          0
KAT             2          2
TEACHER         0          0
MANDELLA        0          0
COWBOY          0          0
BIANCA          3          1
CHASTITY        0          0
WALTER          3          0
SHARON         NA          0
MISS PERKY      0         NA
```

```
Rows: 772 Columns: 14
── Column specification
────────────────────────────────────────────────────────
Delimiter: "\t"
chr (7): Title, IMDB_id, Slug, WeightedDegree, Modularity,
ClusteringCoeffic...
dbl (4): GexfID, ReleaseDate, Edges, Characters
num (3): PathLength, Diameter, ConnectedComponents

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

Movie Specific Examples

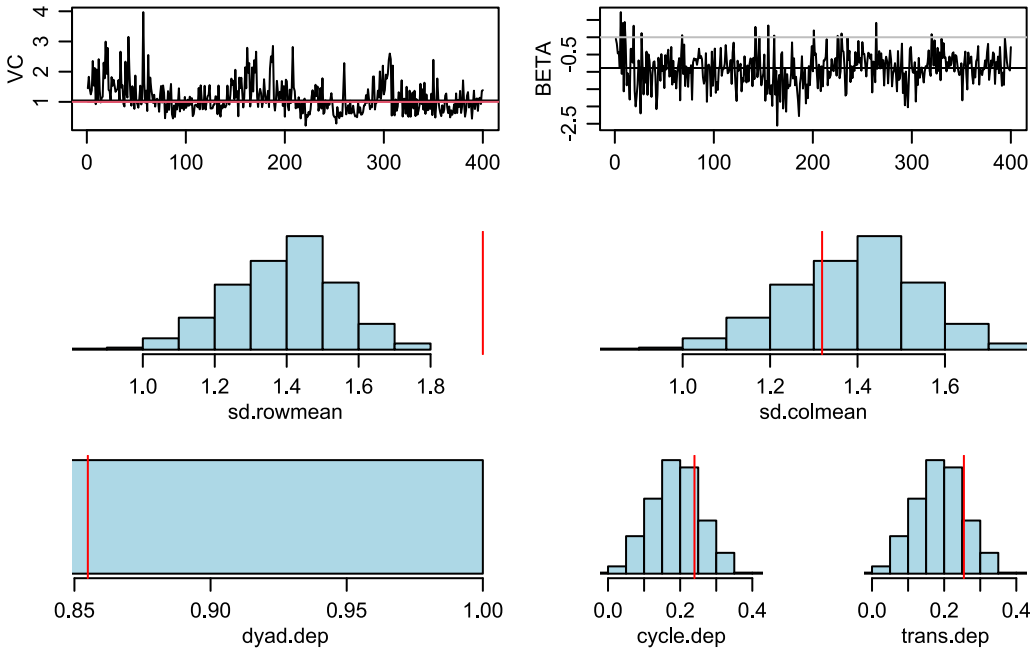
10 Things I Hate About You

Kat, Patrick - Primary love interests

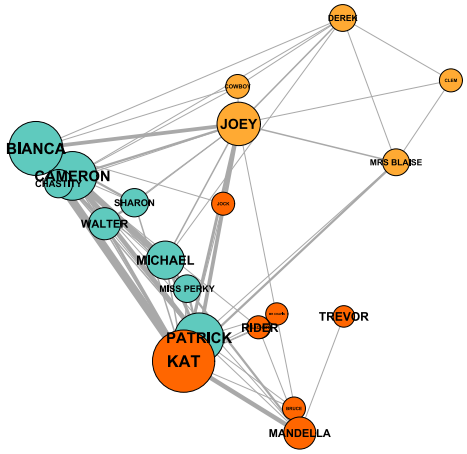
Bianca, Cameron - Secondary love interests

Joey - Main antagonist

Model Plot



Movie Plot

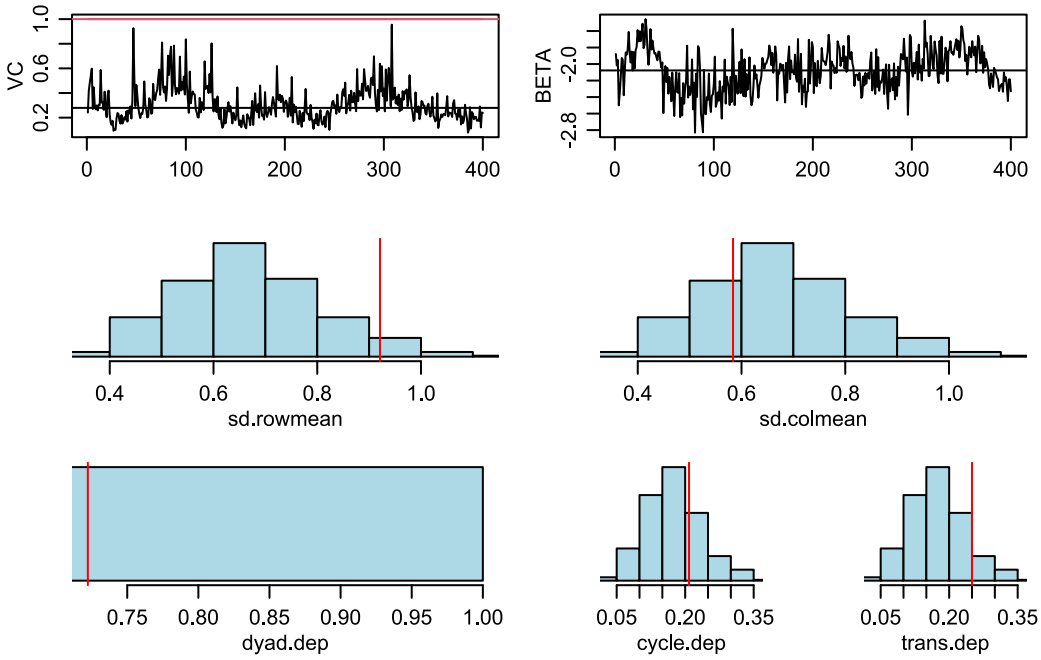


Star Wars

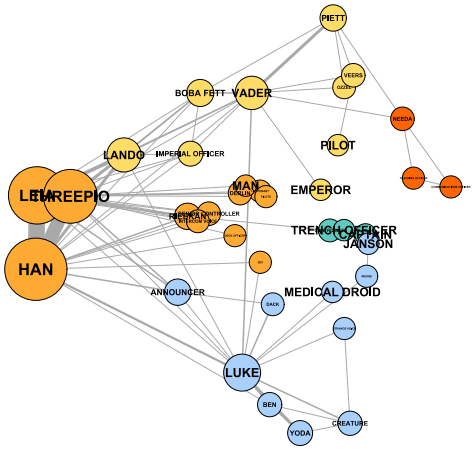
Main characters: Threepio, Leia, Han, Luke

Main Antagonists: Darth Vader, Boba Fett, Emperor

Model Plot



Movie Plot



General Results

Random Movie Sample

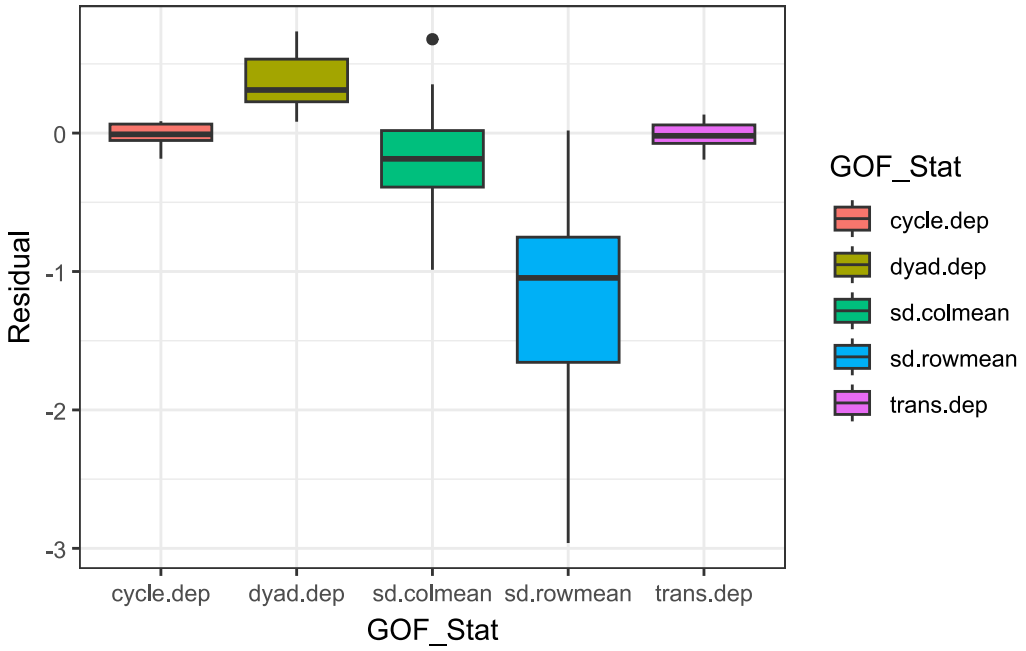
Took random sample of 20 movies and ran same procedure...

```
Warning: The `file` argument of `vroom()` must use `I()` for literal data as
of vroom
1.5.0.

  # Bad:
  vroom("X,Y\n1.5,2.3\n")

  # Good:
  vroom(I("X,Y\n1.5,2.3\n"))
ℹ The deprecated feature was likely used in the readr package.
  Please report the issue at <https://github.com/tidyverse/readr/issues>.
```

```
Rows: 20 Columns: 2
── Column specification
────────────────────────────────────────────────────────
Delimiter: "\t"
chr (1): title
dbl (1): id

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

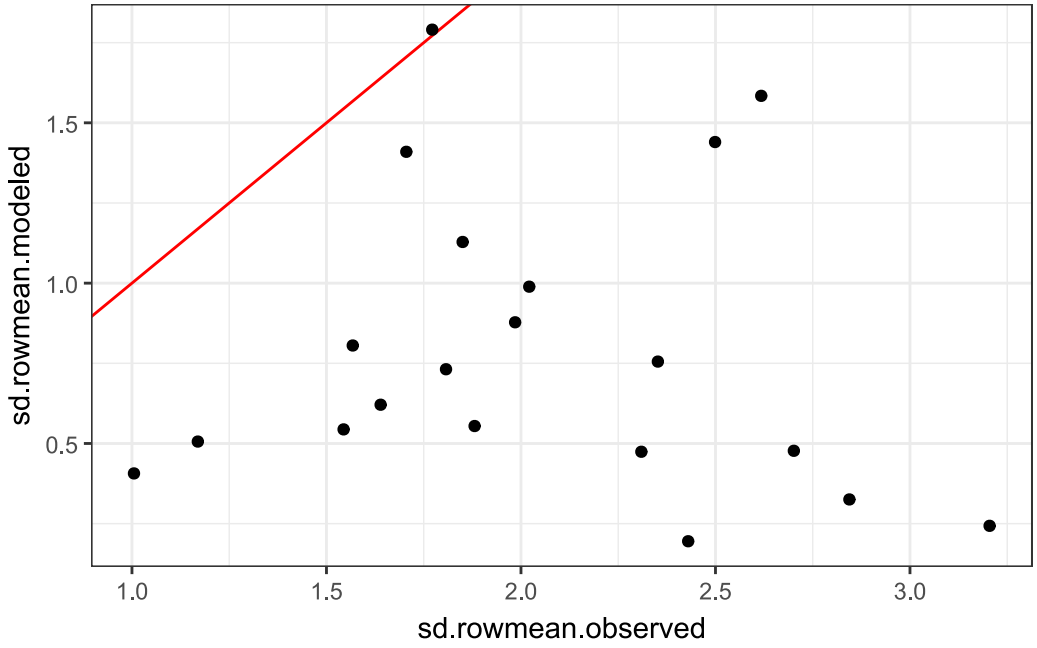
```
 [1] "The Last Samurai"         "Reservoir Dogs"
 [3] "Black Rain"               "Blade Runner"
 [5] "The Reader"               "TRON"
 [7] "Leviathan"                "Alien: Resurrection"
 [9] "Thirteen Days"            "The Crying Game"
[11] "Lord of Illusions"        "Sugar & Spice"
[13] "The Last of the Mohicans" "Crank"
[15] "Being Human"              "The King's Speech"
[17] "Chasing Sleep"            "Death to Smoochy"
[19] "French Kiss"              "Coraline"
```

Goodness of Fits Residuals

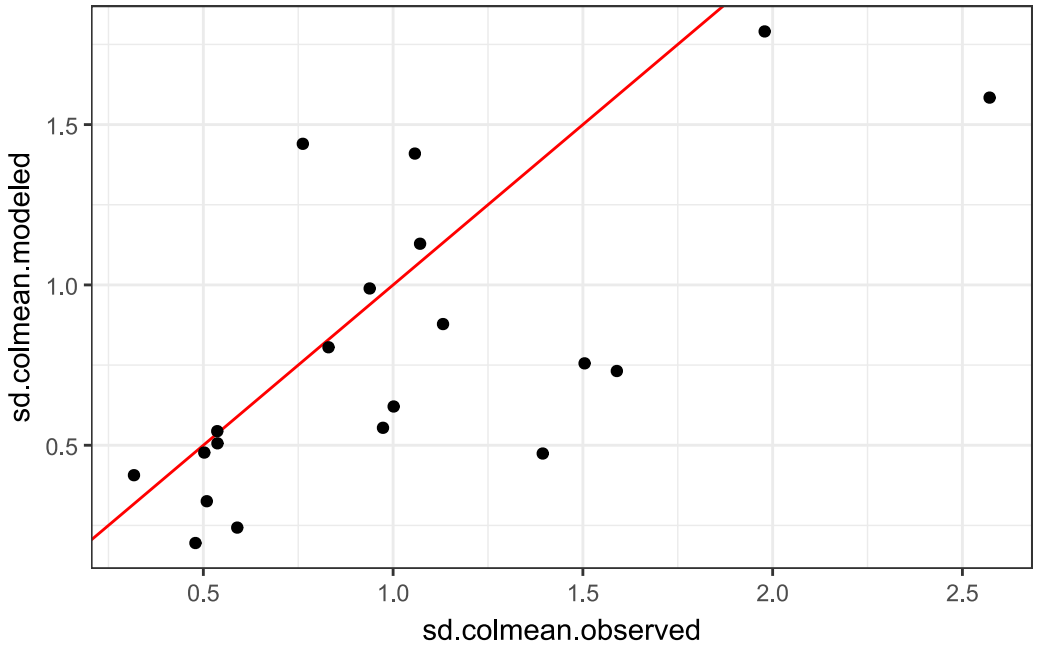


Observed vs Residual

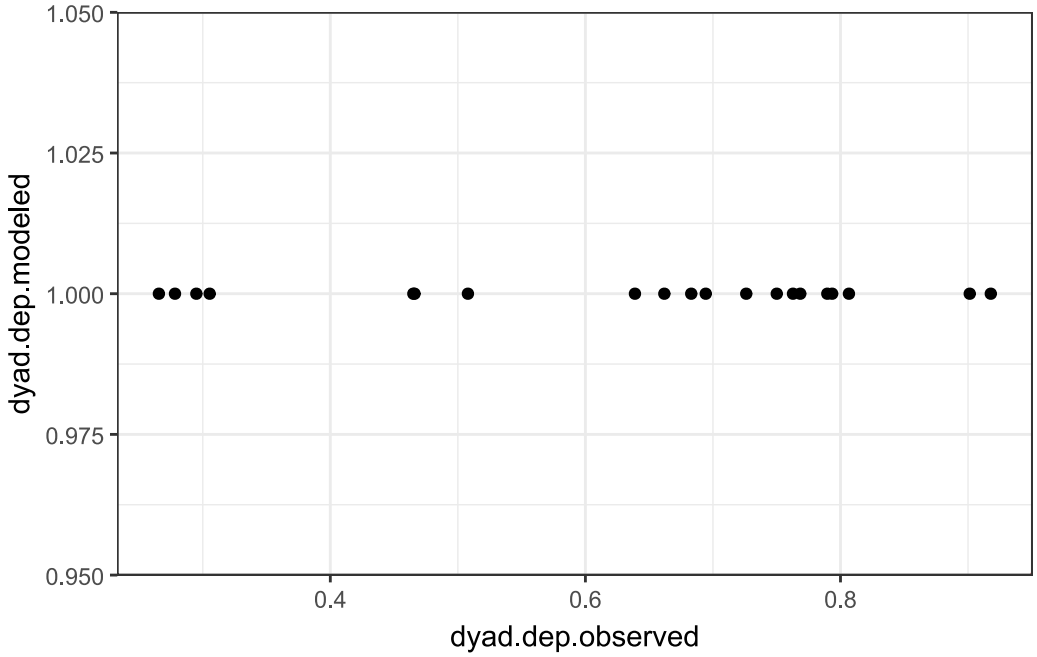
sd(rowmean)



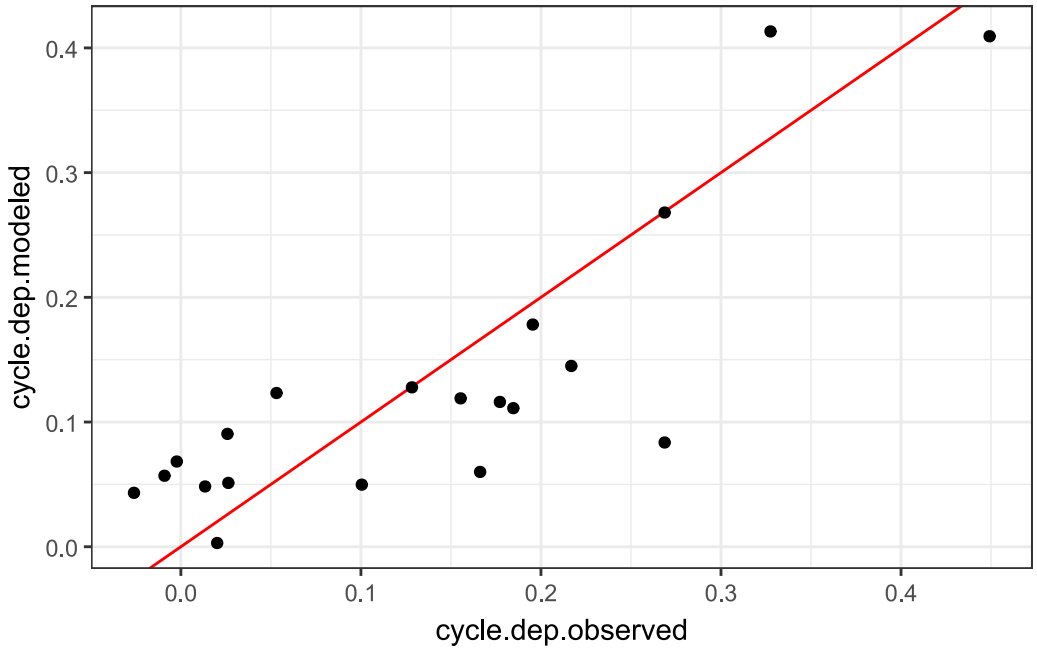
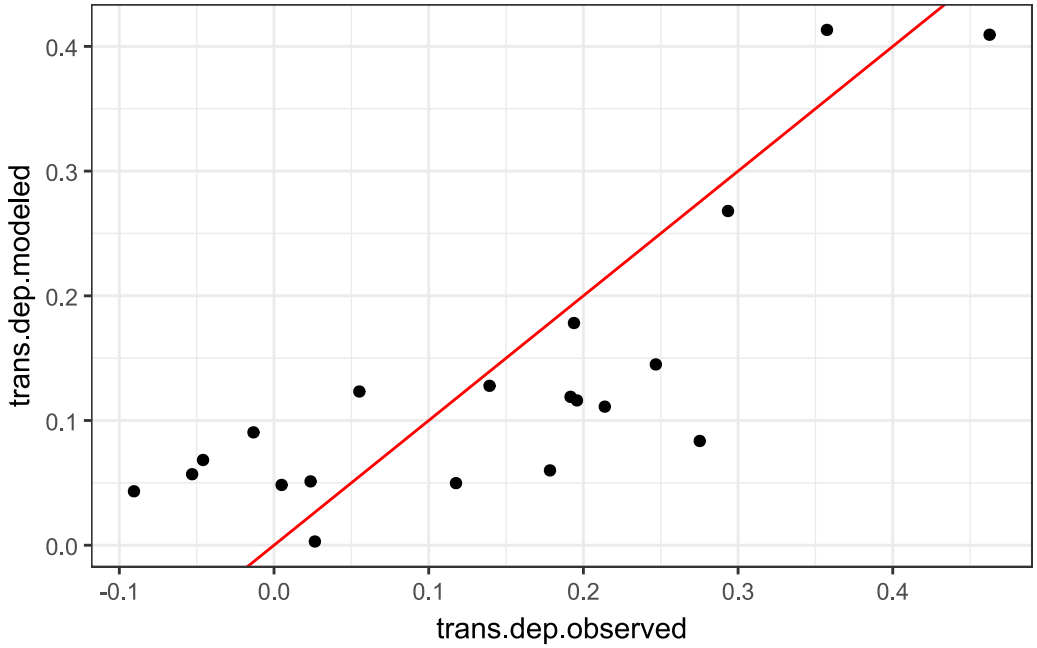
sd(colmean)



dyad.dep



cycle.dep



trans.dep